C.2 Model stability with simulated inference

Figure 2(left) shows the stability of the model predictions under simulated interference. Figure 18 shows the results we obtain when we perform the same analysis using the `Alternate` model variation. The `Large` model was not run due to its high computational load. We find that the `Alternate` model has a larger effect size that `Baseline`, with a $\gtrsim 20\%$ loss of per-token accuracy with only a 1% noise effect. For completeness, we show `QKV-Norm` on the RHS. This is stable by construction, and only jitter due to finite sampling is observed. In these plots, we estimate the statistical uncertainty by evaluating over three datasets and calculating the standard error on the mean. This is plotted as a shaded band, but tends to be narrower than the line width.

Figure 2(right) compares the stability when we only apply noise to sparse heads (defined as $\max_i a_i \geq 95\%$, thin dashed line) and non-sparse heads (defined as $\max_i a_i < 70\%$, thick dashed line). Figure 20 compares these results with the `Alternate` model variation. In both experiments, sparse-attention is stable under %-level noise, and non-sparse distributions dominate this regime.

Note that later layers experience both the artificial noise injection as well as perturbation of their inputs due to the compounding of errors caused by noise in the previous layers.

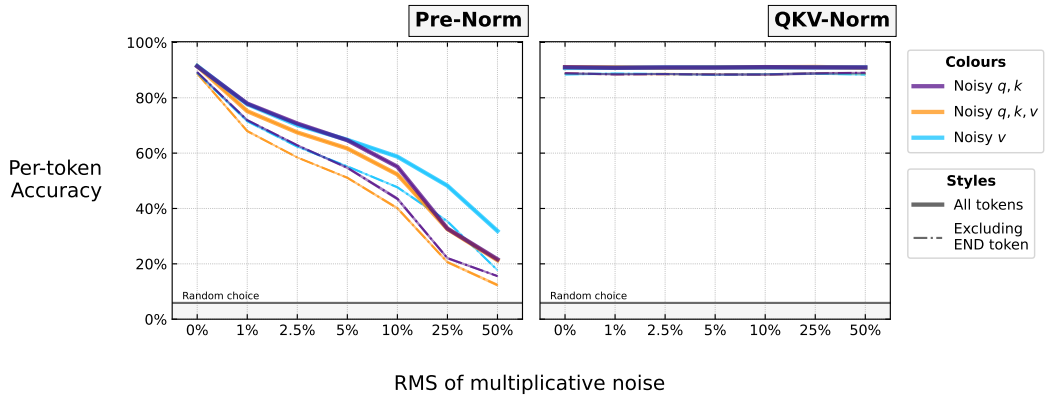

Figure 17: Evolution of per-token accuracy as we increase noise on the L_2-norms of $\{q, k_t, m_t\}$ for the `Baseline` model and task configuration.

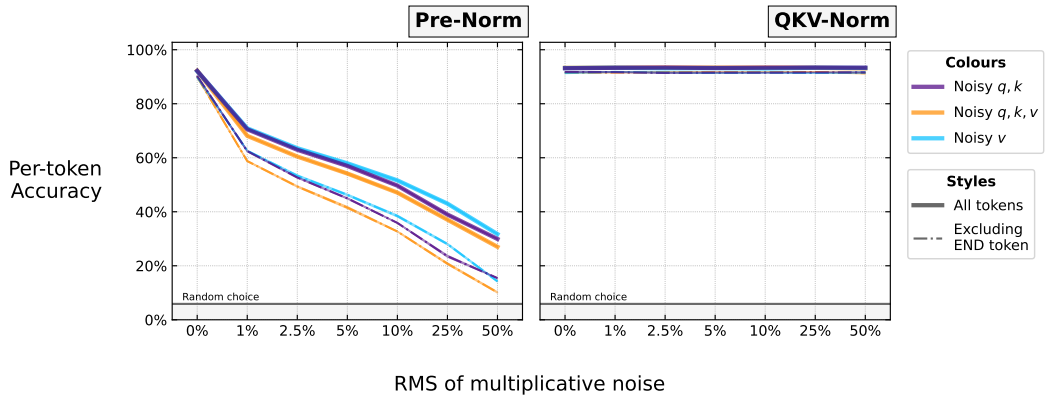

Figure 18: Evolution of per-token accuracy as we increase noise on the L_2-norms of $\{q, k_t, m_t\}$ for the `Alternate` model and task configuration.

C.3 Circuit collapse

Figure 3 shows the probability of circuit collapse. This is the probability that an attention distribution with no noise selects embedding i with high probability $a_i \geq 95\%$, and when noise is added, it transitions such that some $k \neq i$ becomes the maximum attended embedding. Figure 23 shows the results we obtain when we perform the same analysis using the `Alternate` and `Large` model

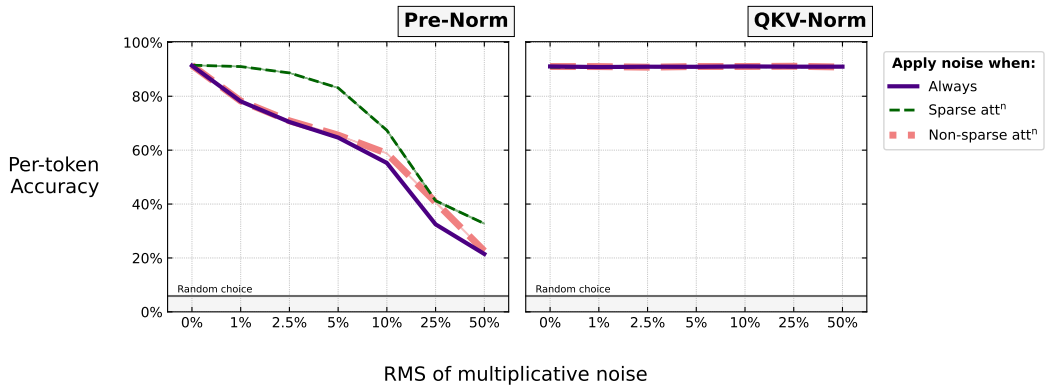

Figure 19: Evolution of per-token accuracy as we increase noise on the L_2-norms of $\{q, k_t, m_t\}$ for the `Baseline` model and task configuration.

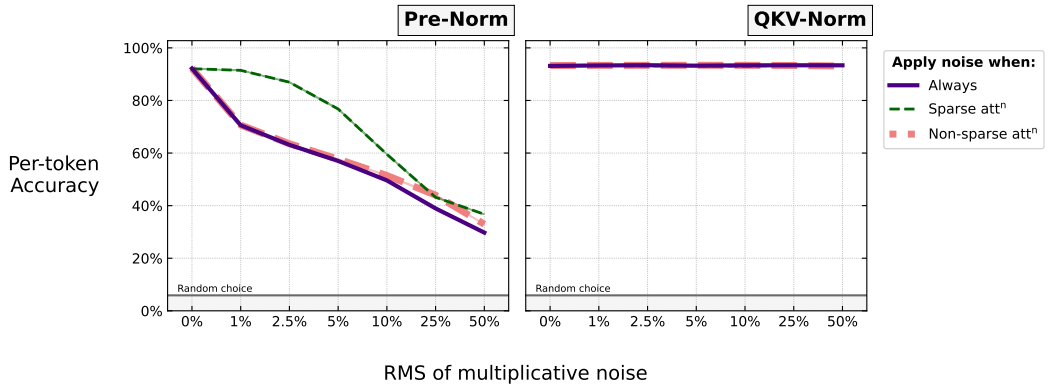

Figure 20: Evolution of per-token accuracy as we increase noise on the L_2-norms of $\{q, k_t, m_t\}$ for the `Alternate` model and task configuration.

variations. In both cases, we observe the onset of circuit collapse at smaller noise levels. Whilst the `Baseline` model observed that 1% of sparse attention heads collapsed with 11% noise, this value is 7.5% for `Alternate` and 5.5% for `Large`.

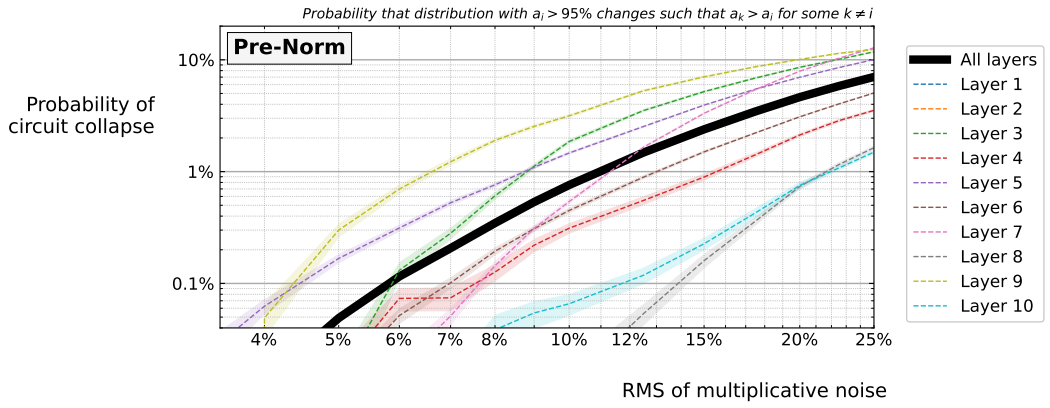

Figure 21: Probability of *circuit collapse* vs increasing noise using the `Baseline` model and task configuration. This is a replication of Figure 3.

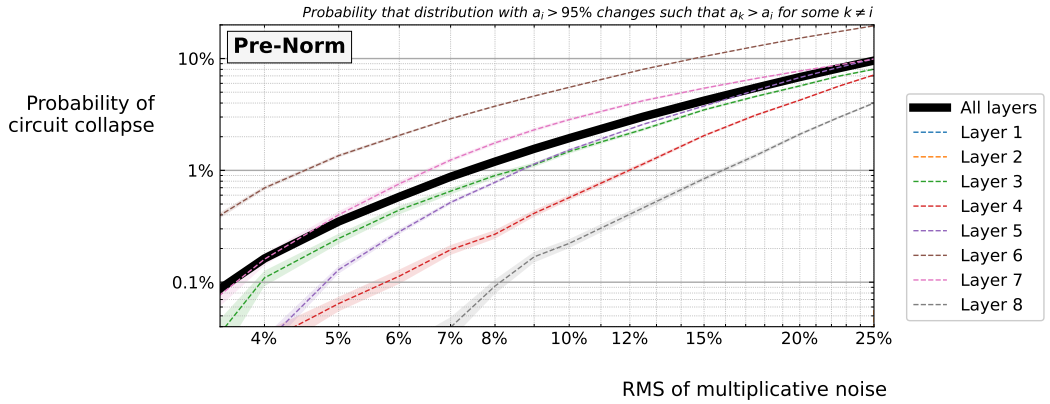

Figure 22: Probability of *circuit collapse* vs increasing noise using the `Alternate` model and task configuration.

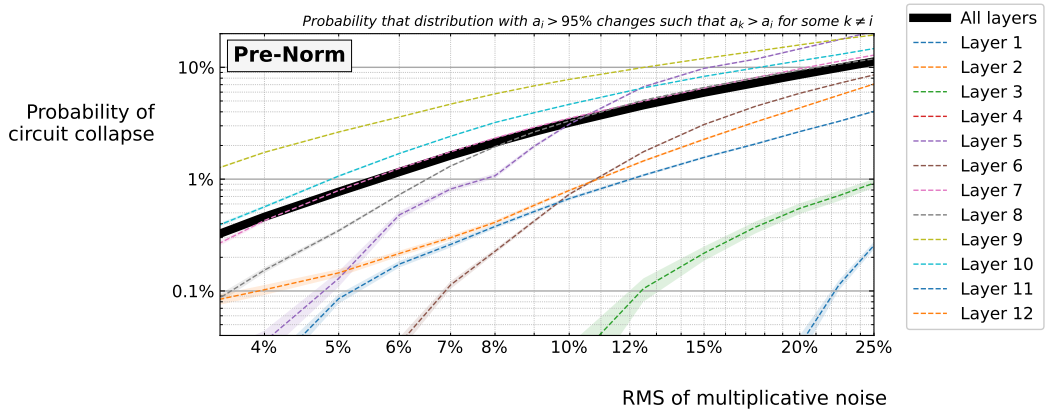

Figure 23: Probability of *circuit collapse* vs increasing noise using the `Large` model and task configuration.